Project Idea: Game Library & Backlog Tracker

1. Project Description

I want to develop an application that allows users to manage their video game collections. Users can register and log in to organize their personal game libraries. They can add games and set their current progress status, such as "Not Started", "Playing", "Completed", or "Dropped". Additionally, users can rate games they have played. Each game is assigned to a genre to make the collection easier to organize. The application aims to help users keep track of games they have already played and those they still want to play.

2. Entities

The application will require four database entities.

ApplicationUser

Stores user information.

- id: Long
- username: String
- passwordHash: String

Game

Stores information about video games.

- id: Long
- title: String
- description: String
- releaseDate: LocalDate
- genreId: Long (Foreign Key)

Genre

Stores different video game genres.

- id: Long
- name: String

LibraryEntry

Connects a user to a game and stores their personal progress.

- id: Long
- userId: Long (Foreign Key)
- gameId: Long (Foreign Key)
- status: String
- rating: Integer (1–10)

Relationships

- ApplicationUser → LibraryEntry (1): One user can have multiple games in their library.
- Game → LibraryEntry (1): One game can appear in multiple users' libraries.
- Genre → Game (1): One genre can contain multiple games, while each game belongs to one genre.

The LibraryEntry entity creates a many-to-many (N) relationship between ApplicationUser and Game.

3. Motivation

I chose this project idea because I am interested in video games and want to keep track of the games I have played. Over time, it can become difficult to remember which games I have already completed and which ones I still want to play. With this application, I could organize my personal game library and track my progress. At the same time, this project will allow me to improve my skills in Java, Quarkus, REST APIs, databases, and JWT authentication.